

## DRY, YAGNI, and Clean Code principles

```
#include <iostream>

#include <string>

using namespace std;

string calculateGrade(int marks) {
    if (marks >= 90)
        return "Grade A";
    if (marks >= 80)
        return "Grade B";
    if (marks >= 70)
        return "Grade C";
    if (marks >= 60)
        return "Grade D";
    return "Fail";
}

string checkEligibility(int attendance) {
    return (attendance < 75) ? "Not eligible" : "Eligible";
}

void handleStudent() {
    int marks, attendance;

    cout << "Enter marks: ";

    cin >> marks;

    cout << calculateGrade(marks) << endl;

    cout << "Enter attendance: ";

    cin >> attendance;

    cout << checkEligibility(attendance) << endl;
}

void handleTeacher() {
```

```

int salary, yearsOfExperience;

cout << "Enter salary: ";

cin >> salary;

cout << (salary > 50000 ? "High salary" : "Low salary") << endl;


cout << "Enter years of experience: ";

cin >> yearsOfExperience;

cout << (yearsOfExperience > 10 ? "Senior Teacher" : "Junior Teacher") << endl;
}


int main() {
    int choice;

    cout << "Enter choice (1 for Student, 2 for Teacher): ";

    cin >> choice;


    switch (choice) {
        case 1:
            handleStudent();

            break;

        case 2:
            handleTeacher();

            break;

        default:
            cout << "Invalid choice." << endl;

            return 0;
    }


    cout << "Done" << endl;

    return 0;
}

```

## Folder structure using Domain-Driven Design (DDD) for a Hospital Management System

```
Session Actions Edit View Help
(kaium@4g0t)~[~/Desktop]app
$ tree HospitalManagementSystem
HospitalManagementSystem
├── HospitalManagementSystem
│   ├── API
│   │   ├── Controllers
│   │   │   ├── AppointmentsController.cs
│   │   │   ├── PatientsController.cs
│   │   │   └── Program.cs
│   │   ├── Application
│   │   │   ├── Appoinments
│   │   │   │   ├── ScheduleAppointment
│   │   │   │   │   ├── ScheduleAppointmentCommand.cs
│   │   │   │   │   └── ScheduleAppointmentHandler.cs
│   │   │   ├── Interfaces
│   │   │   │   └── IEventBus.cs
│   │   │   ├── Patients
│   │   │   │   ├── GetPatient
│   │   │   │   │   ├── GetPatientHandler.cs
│   │   │   │   │   └── GetPatientQuery.cs
│   │   │   │   ├── RegisterPatient
│   │   │   │   │   ├── RegisterPatientCommand.cs
│   │   │   │   │   └── RegisterPatientHandler.cs
│   │   └── Domain
│   │       ├── Appoinments
│   │       │   ├── Appoinment.cs
│   │       │   ├── AppointmentId.cs
│   │       │   ├── Events
│   │       │   │   └── AppointmentScheduledEvent.cs
│   │       │   └── IAppointmentRepository.cs
│   │       ├── Patients
│   │       │   ├── Events
│   │       │   │   └── PatientRegisteredEvent.cs
│   │       │   ├── IPatientRepository.cs
│   │       │   ├── Patient.cs
│   │       │   ├── PatientId.cs
│   │       └── Shared
│   │           ├── BaseEntity.cs
│   │           ├── DomainException.cs
│   │           ├── ValueObjects
│   │           │   ├── Address.cs
│   │           │   └── Money.cs
│   └── Infrastructure
│       ├── EventBus
│       │   └── RabbitMQEventBus.cs
│       └── Persistence
│           ├── ApplicationDbContext.cs
│           ├── AppointmentRepository.cs
│           └── PatientRepository.cs
└── tests

22 directories, 26 files
(kaium@4g0t)~[~/Desktop]
```